

A Hybrid Deep Learning Framework for Accurate Jute Pest Detection

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ABSTRACT Pest infestation in jute crops significantly reduces yield and quality, posing a serious challenge for farmers and agricultural productivity [1]. Traditional pest detection methods are time-consuming and error-prone, making automation essential [1], [2]. The research develops an accurate jute pest classification system through a deep learning framework using convolutional neural networks and vision transformers. We evaluate three pre-trained convolutional models—DenseNet201, EfficientNet, and InceptionV3—trained in both optimized and raw forms to understand the impact of input diversity. [3]–[5]. To further enhance performance, we introduce hybrid models that integrate each CNN with a Vision Transformer, combining local feature extraction with global context understanding [6]. All models are trained and validated on the Jute Pest Dataset, which contains images from 17 different pest classes and 1 category from Agricultural Pests Image Dataset [7], [8]. The system performance evaluation depends on accuracy metrics and precision in addition to recall and F1-score metrics. Results show that optimized models outperform raw models, and hybrid architectures provide additional improvements in classification accuracy and robustness. The research shows how deep learning and transformer-based models can be used to create effective and precise detection systems that track pests in agricultural environments [9]. The proposed system helps farmers and researchers identify pests early, which enables prompt action for sustainable crop management.

INDEX TERMS Jute pest detection, convolutional neural networks, data augmentation, deep learning, DenseNet201, EfficientNet, hybrid models, image classification, InceptionV3, transfer learning, Vision Transformer

I. INTRODUCTION

JUTE is one of the most widely cultivated natural fiber crops, essential for sustainable products like eco-friendly packaging and textiles [1], [10]. The production of jute remains problematic because pests cause major adverse effects on both product quality and quantity [1]. The detection methods employed for pests rely on human observation but prove inconsistent and time-consuming since they are error-prone [2]. Therefore, systematic pest detection system with automated functions and reliable performance needs development to benefit jute farmers and agricultural researchers.

Image classification breakthroughs emerge because of deep learning progress of recent times [11]. Convolutional neural networks (CNNs) such as DenseNet201, EfficientNet, and InceptionV3 are widely used for extracting spatial features from images [3]–[5]. Meanwhile, Vision Transformers (ViTs) have recently shown strong performance in capturing global dependencies using attention mechanisms. In this work, we investigate both CNN-based and hybrid CNN+ViT

architectures to detect and classify jute pests accurately from image data [6].

We train six models: three standalone CNN models in both raw and optimized forms, and three hybrid models combining each CNN with a Vision Transformer. These models are evaluated on a labeled dataset containing images from 18 pest classes, including a newly added Grasshopper category.

A. ABBREVIATIONS AND ACRONYMS

We use the following abbreviations in this article:

- CNN – Convolutional Neural Network
- ViT – Vision Transformer
- F1 Score – Harmonic mean of precision and recall
- GPU – Graphics Processing Unit

II. RELATED WORK

Deep learning systems for automated pest detection have become major interests in current years due to their role in agricultural sustainability and yield protection [9], [12].

Traditional pest identification methods involve manual inspection and expert analysis, which are time-consuming, subjective, and not scalable for large-scale farming [2]. The combination of convolutional neural networks (CNNs) together with transformers through deep learning and computer vision technology has delivered high-quality performance in identifying pests. [11].

Li [13] proposed a YOLO-based model called YOLO-JD for detecting jute pests and diseases. Their approach employed object detection to identify pest instances in real-time scenarios and demonstrated strong performance on custom datasets. However, the model was more suitable for detection than fine-grained classification.

In a more classification-focused study, Talukder [2] developed JutePestDetect, a transfer learning-based framework for classifying jute pests. They fine-tuned pre-trained CNNs such as ResNet50 and DenseNet121 and reported high accuracy using a dataset containing multiple jute pest categories [2]. Their results validated the potential of transfer learning but did not explore hybrid modeling.

Rahman and Islam [14] also investigated transfer learning for pest classification. Their work highlighted training efficiency and high accuracy using lightweight architectures. While their method achieved good results, it did not incorporate transformer models or evaluate hybrid frameworks.

Although CNNs have demonstrated excellent spatial feature extraction, they are limited in modeling long-range dependencies. Vision Transformers (ViTs), based on self-attention mechanisms, overcome this limitation and are capable of capturing global context from input images. However, the use of ViTs in pest detection, particularly for crops like jute, remains relatively unexplored [6].

In this project, we extend previous work by implementing both CNN-based and hybrid CNN+ViT models. We evaluate three popular CNN architectures—DenseNet201, EfficientNet, and InceptionV3—individually and in combination with Vision Transformers. Additionally, we compare the performance of optimized and raw models. Our contribution highlights that hybrid architectures, which integrate both local and global feature representations, can surpass conventional CNNs in accurately classifying diverse jute pest species. [9].

III. DATASET OVERVIEW

The experimental evaluation conducted in this study is based on a curated dataset derived primarily from the Jute Pest Dataset available on the UCI Machine Learning Repository [7]. The dataset originally comprised 17 pest categories. To enhance its diversity and increase its applicability in real-world pest classification scenarios, we incorporated an additional class—Grasshopper—from a publicly available external dataset on Kaggle [8]. With this integration, the final dataset consists of a total of 7,721 high-resolution images spread across 18 distinct jute pest categories. Supervised image classification works best with this data because the organized folders contain each class's information in separate labeled directories [2].



FIGURE 1. Sample images from 6 different jute pest categories with 3 examples per class. These samples illustrate the diversity and visual complexity of the dataset.

The complete dataset was divided into three parts:

- **Training set:** 89.15%
- **Validation set:** 5.66%
- **Test set:** 5.19%

This split was intentionally skewed toward training to maximize model exposure to data while still reserving sufficient samples for validation and testing [12].

The image distribution among classes shows unevenness since the Jute Stem Weevil along with Jute Aphid classes possess greater numbers of images compared to Beet Armyworm and Termite classes. Training algorithms require effective preprocessing and augmentation techniques because the current class distribution shows an imbalance.

Figure 2 demonstrates how training, validation and test splits contain images distributed among different classes. Figure 3 shows the percentage-wise distribution of each class within the dataset.

These visualizations help assess the overall class balance and inform decisions related to model training, augmentation, and evaluation strategies.

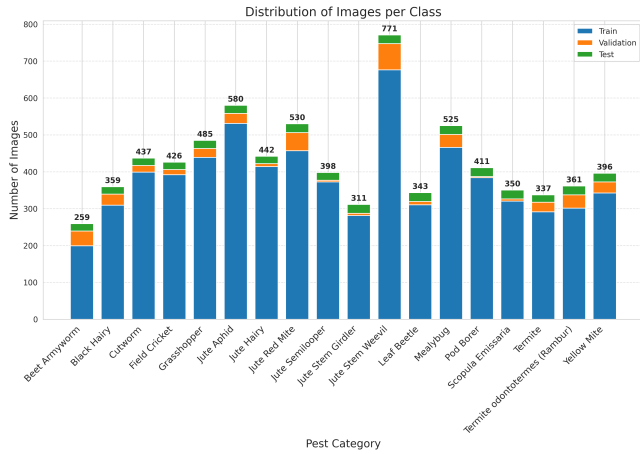


FIGURE 2. Distribution of images per class in the Jute Pest Dataset across training, validation, and test sets.

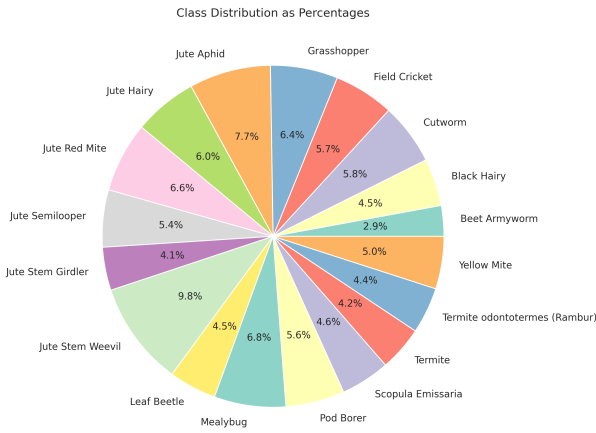


FIGURE 3. Class distribution as percentages. This highlights imbalance across pest categories.

IV. METHODOLOGY

This section outlines the preprocessing pipeline, deep learning models implemented, and the training procedures applied to perform pest classification in jute crops.

A. DATA PREPROCESSING

The data preprocessing pipelines were designed to align with the training objectives of different model types. Images were resized according to the input size requirements of each model architecture: DenseNet201 and EfficientNetV2 used a standard dimension of 224×224 pixels, while InceptionV3 required a dimension of 299×299 pixels [3], [4]. Pixel values were normalized using ImageNet statistics (mean: [0.485, 0.456, 0.406], std: [0.229, 0.224, 0.225]). Beyond this, the preprocessing varied slightly depending on whether the model was a standard CNN or a hybrid CNN+Vision Transformer (ViT).

1) Optimized CNN Models

The CNN models—DenseNet201, EfficientNetV2-S, and InceptionV3—were trained in their optimized forms using the following transformation pipeline:

- **Resize:**
 - For DenseNet201 / EfficientNetV2: Scales the image to 224 pixels on the shorter edge.
 - For InceptionV3: Scales the image to 299 pixels on the shorter edge.
- **RandomResizedCrop:**
 - For DenseNet201 / EfficientNetV2: Random crop to 224×224 with a scale range between 0.5 and 1.0.
 - For InceptionV3: Random crop to 299×299 with a scale range between 0.5 and 1.0.
- **RandomHorizontalFlip:** Flips the picture horizontally with a 0.5 chance.
- **RandomRotation:** Rotates the picture at random by up to $\pm 15^\circ$.
- **ColorJitter:** Randomly changes brightness, contrast, saturation, and hue.
- **ToTensor:** Converts the image into a PyTorch tensor.
- **Normalize:** Applies ImageNet normalization (mean: [0.485, 0.456, 0.406], std: [0.229, 0.224, 0.225]).

Additionally, two advanced regularization techniques—Mixup and CutMix—were applied to CNN models during training to improve generalization and mitigate overfitting, particularly for underrepresented classes [15], [16].

- **Mixup Augmentation:** Mixup creates synthetic training examples by linearly interpolating between pairs of input images and their respective labels.

$$\tilde{x} = \lambda x_i + (1 - \lambda) x_j \quad (1)$$

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j \quad (2)$$

where x_i and x_j are input images, y_i and y_j are the associated one-hot encoded labels, and $\lambda \sim \text{Beta}(\alpha, \alpha)$ is sampled from a Beta distribution.

- **CutMix Augmentation:** CutMix improves diversity by replacing a rectangular piece of one image with another and combining labels based on patch area.

Image Patch Mixing:

$$\tilde{x} = x_i \odot M + x_j \odot (1 - M) \quad (3)$$

Label Mixing:

$$\tilde{y} = \lambda y_i + (1 - \lambda) y_j \quad (4)$$

where M is a binary mask that defines the inserted patch region, and λ is computed as:

$$\lambda = 1 - \frac{A_{\text{patch}}}{A_{\text{image}}} \quad (5)$$

Here, A_{patch} denotes the area of the cut region, and A_{image} is the total area of the image.

The utilized augmentation techniques enhanced the network robustness because they improved both generalization ability and minimized overfitting across different pest categories [15], [16].

2) Hybrid CNN + ViT Models

For the hybrid models (DenseNet201 + ViT, EfficientNetV2-S + ViT, and InceptionV3 + ViT), a simpler preprocessing pipeline was used, focused on preserving structural details without introducing excessive variability:

- **Resize:**
 - For DenseNet201 / EfficientNetV2: Resizes the image to 224 pixels on the shorter side.
 - For InceptionV3: Resizes the image to 299 pixels on the shorter side.
- **CenterCrop:**
 - For DenseNet201 / EfficientNetV2: Crops the center 224×224 region of the image.
 - For InceptionV3: Crops the center 299×299 region of the image.
- **ToTensor:** Converts the image to a tensor.
- **Normalize:** Applies ImageNet normalization (mean: [0.485, 0.456, 0.406], std: [0.229, 0.224, 0.225]).

Since Vision Transformers are sensitive to positional and structural integrity, aggressive augmentations were avoided. This helped maintain consistency in token embedding across different spatial arrangements.

Each preprocessing approach was tuned based on the model type, striking a compromise between data variability and feature fidelity. [12].

B. MODEL ARCHITECTURES

In this study, we explore two categories of models for jute pest classification: (1) standalone convolutional neural networks (CNNs) and (2) hybrid models combining CNNs with Vision Transformers (ViTs). Each architecture was selected for its proven effectiveness in image classification tasks and fine-tuned to detect and differentiate between 18 pest categories in the dataset [2], [6].

1) CNN Models

We fine-tuned three pre-trained CNN architectures: DenseNet201, EfficientNetV2-S, and InceptionV3. These models are widely recognized for their strong performance in large-scale image classification and were adapted to the specific requirements of jute pest recognition [3]–[5].

- **DenseNet201:** DenseNet, or Densely Connected Convolutional Network, introduces direct connections between all layers in a feed-forward manner. Each layer receives inputs from all the previous layers, enhancing feature reuse and reducing vanishing gradient issues. DenseNet201, with its deep structure and dense connectivity, is particularly useful for capturing fine-grained

features in pest images, especially in categories with visual similarities [3].

- **EfficientNetV2-S:** EfficientNet uses a compound scaling approach to equally alter network depth, breadth, and input resolution with predetermined scaling coefficients. The V2-S version balances performance and efficiency, making it well-suited for deployment in resource-constrained applications while retaining competitive accuracy. [5].
- **InceptionV3:** InceptionV3 uses factorized convolutions and dimensionality reduction techniques within “Inception modules” to increase depth and width without excessive computational cost. It processes visual information at multiple scales simultaneously, which is beneficial for identifying pests with varying textures and shapes in diverse environmental conditions.

All three CNN models were trained under two settings: raw and optimized. The raw training setting assessed the models’ baseline performance, while the optimized setting—enhanced through data augmentation—aimed to improve generalization and robustness against variations in image conditions. [12], [16].

2) Hybrid Models with Vision Transformers

While CNNs are excellent at extracting local features, they are limited in modeling long-range dependencies. To address this, we also developed hybrid models that integrate CNN backbones with Vision Transformers (ViTs). ViTs work by considering pictures as sequences of patches and using self-attention to model global relationships across the entire image. [17].

In our hybrid approach, the feature maps generated by the CNNs are flattened and projected into a form compatible with the ViT encoder. The combined architecture thus captures both detailed spatial information and high-level contextual patterns [6].

The hybrid models explored are:

- **DenseNet201 + ViT:** Combines the detailed local feature extraction of DenseNet with the global context modeling of ViT, ideal for distinguishing pests with subtle visual differences.
- **EfficientNetV2-S + ViT:** Merges a lightweight yet powerful CNN backbone with ViT’s attention mechanism, offering an efficient yet accurate solution for complex classification.
- **InceptionV3 + ViT:** Leverages InceptionV3’s multi-scale analysis with ViT’s global perception, enhancing recognition in visually cluttered or heterogeneous images.

These hybrid models were designed to maximize feature richness and capture semantic patterns beyond what is possible with CNNs alone. Their performance is particularly valuable in scenarios with complex backgrounds or similar pest appearances [6].

C. TRAINING PROCESS

All models were implemented using PyTorch and trained using GPU acceleration on the Kaggle platform. The training pipeline was designed to ensure robustness and generalizability using best practices in deep learning for classification tasks [11].

- **Optimizer:** Adam optimizer was used for all models, with an initial learning rate of 0.00005 [18].
- **Loss Function:** Weighted CrossEntropyLoss was applied to counter class imbalance. Specific classes such as Jute Hairy, Cutworm, Beet Armyworm, and Mealybug were given higher weights [19].

The categorical cross-entropy loss function:

$$\mathcal{L} = - \sum_{i=1}^N y_i \log(\hat{y}_i) \quad (6)$$

where y_i is the true label, $\hat{y}_i$ is the predicted probability for class i , and N is the number of classes.

- **Learning Rate Scheduler:** ReduceLROnPlateau was utilized to lower the learning rate when validation loss plateaued, resulting in better convergence.
- **Epochs:** Each model was trained for a maximum of 10 epochs.
- **Batch Size:** The training and validation processes operated with a batch size of 64.
- **Early Stopping:** Triggered after 5 consecutive epochs with no improvement in validation accuracy.

Training Procedure: At each epoch:

- The model was trained using augmented data and class-weighted loss.
- Validation was performed on a separate set with standard preprocessing (no augmentation).
- Validation accuracy and loss were logged, and learning rate was adjusted accordingly.
- The best model (based on validation accuracy) was saved.

D. EVALUATION METRICS:

To assess the performance of each model, we employed a range of widely-used evaluation metrics suitable for multi-class classification tasks:

- **Accuracy:** The proportion of correctly predicted instances among all predictions. While useful for measuring overall performance, accuracy alone is insufficient for imbalanced datasets [19].

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \quad (7)$$

- **Precision:** The ratio of true positives to total predicted positives. It evaluates how well the model avoids false positives. [19].
- **Recall:** The ratio of true positives to the number of positives in the dataset. It demonstrates the model's ability to identify all relevant cases (true positives). [19].

- **F1 Score:** When dealing with class-imbalanced datasets the harmonic mean between precision and recall offers a balanced evaluation approach. Macro-averaged F1 scores were used for overall comparison [19], [20].

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

- **Confusion Matrix:** The confusion matrix measures the model's performance by comparing predicted and actual labels. It demonstrates the model's ability to distinguish between pest classes and identifies common misclassification. [19].

V. RESULTS AND EVALUATION

This section presents and compares the performance of both standalone CNN models and hybrid CNN+ViT models using key evaluation metrics. Accuracy, F1-score, and validation accuracy trends are analyzed across different architectures.

A. RAW VS. OPTIMIZED CNN MODELS

The bar chart in Fig. 4 illustrates the test accuracy of three CNN models—InceptionV3, DenseNet201, and EfficientNetV2-S—trained in both raw and optimized forms. A clear improvement is observed across all three models in their optimized versions. The highest test accuracy of 97.8% was achieved by the optimized InceptionV3 model [12], [16].

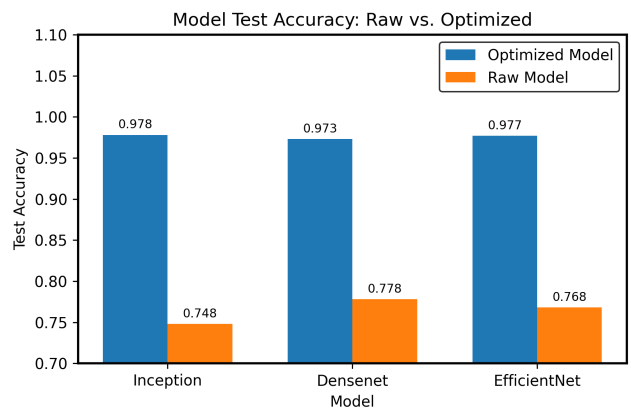


FIGURE 4. Test accuracy comparison between raw and optimized CNN models.

B. VALIDATION ACCURACY OF HYBRID MODELS

The line graph in Fig. 5 shows the validation accuracy over 10 epochs for three hybrid models: DenseNet201+ViT, EfficientNet+ViT, and Inception+ViT. All models show rapid convergence, with Inception+ViT and DenseNet201+ViT consistently outperforming others. Inception+ViT achieves the highest validation accuracy, indicating strong generalization during training [6].

C. F1 SCORE ANALYSIS OF HYBRID MODELS

Table 1 presents the F1-score for each pest class across the three hybrid models: DenseNet201 + ViT, EfficientNetV2-S

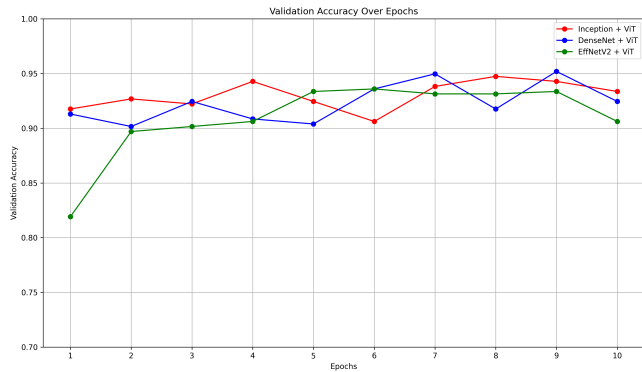


FIGURE 5. Validation accuracy across epochs for the hybrid models.

+ ViT, and InceptionV3 + ViT. The results clearly show that most classes achieved an F1-score of 0.95 or higher across all models, indicating strong classification performance [6], [19].

TABLE 1. Class-wise F1 Scores for Hybrid Models

Category	DenseNet + ViT	EfficientNet + ViT	Inception + ViT
Beet Armyworm	0.92	0.88	0.97
Black Hairy	1.00	0.97	1.00
Cutworm	0.95	0.86	0.98
Field Cricket	0.98	0.98	1.00
Grasshopper	0.95	0.93	1.00
Jute Aphid	1.00	0.96	1.00
Jute Hairy	0.98	0.97	0.98
Jute Red Mite	1.00	0.98	0.96
Jute Semilooper	0.98	0.98	0.98
Jute Stem Girdler	1.00	1.00	1.00
Jute Stem Weevil	1.00	1.00	1.00
Leaf Beetle	1.00	1.00	1.00
Mealybug	1.00	0.93	1.00
Pod Borer	1.00	1.00	0.96
Scopula Emissaria	1.00	1.00	1.00
Termite	1.00	1.00	1.00
Termite Odontotermes	1.00	1.00	1.00
Yellow Mite	1.00	0.91	1.00
Average	0.987	0.964	0.990

As evident from the results, Inception + ViT achieved the highest average F1-score of 0.990, followed closely by DenseNet201 + ViT at 0.987. EfficientNetV2-S + ViT, while slightly lower at 0.964, still maintained strong class-wise performance and offers computational efficiency for real-time deployment [6], [19]. These findings further validate the hybrid models' ability to generalize effectively across all 18 pest categories.

D. COMPARISON WITH EXISTING WORKS

To assess the significance of our approach, we compare its performance against various state-of-the-art models previously applied to the Jute Pest Dataset [2], [10], [13]. Table 2 shows a summary of these existing models, the datasets used, and their reported accuracies, along with the results of our hybrid CNN + ViT model [6].

As shown in Table 2, Our Hybrid CNN + ViT model achieves the highest reported accuracy of 99% on the Dataset (Combining jute Pest dataset and Agricultural Pests Image Dataset dataset), surpassing all previously published approaches [6], [10], [13]. This result highlights the effectiveness of integrating convolutional neural networks for local

TABLE 2. Summary of Different Studies and Their Accuracies Compared to our Model

Reference	Model	Dataset	Accuracy
Li et al. [13]	YOLO-JD	Jute Pest Dataset	Not mentioned
Talukder et al. [2]	JutePestDetect	Jute Pest Dataset	97.8%
Rahman & Islam [14]	Transfer Learning	Jute Pest Dataset	96.5%
Rahman & Islam [14]	ResNet50	Jute Pest Dataset	96.3%
Rahman & Islam [14]	VGG16 (Pretrained)	Jute Pest Dataset	93.7%
Rahman & Islam [14]	MobileNetV2	Jute Pest Dataset	91.8%
Rahman & Islam [14]	DenseNet201	Jute Pest Dataset	92.9%
Rahman & Islam [14]	EfficientNet-B3	Jute Pest Dataset	97.1%
Rahman & Islam [14]	InceptionV3	Jute Pest Dataset	96.0%
Rahman & Islam [14]	Xception	Jute Pest Dataset	95.5%
Rahman & Islam [14]	Hybrid CNN-Transformer	Jute Pest Dataset	98.1%
Our Model	Hybrid CNN + ViT	Our Dataset	99%

feature extraction with Vision Transformers for capturing global context [6].

The superior performance also reflects the positive impact of advanced data augmentation, balanced preprocessing, and the strategic fusion of model architectures. These improvements enable the proposed model to better handle class imbalance, varied visual patterns, and subtle pest distinctions, ultimately making it more suitable for practical deployment in real-world agricultural scenarios [6].

VI. CONCLUSION

- All models utilize a CNN backbone, and hybrid models additionally incorporate a Vision Transformer (ViT) to combine local feature extraction with global context understanding [6].
- CNN models (InceptionV3, DenseNet201, EfficientNetV2-S) demonstrated strong baseline performance, with notable improvements in their optimized forms. InceptionV3, when trained as an optimized model with augmentation, achieved a test accuracy of 97.8%, showcasing the enduring effectiveness of augmentation-based preprocessing [12].
- Hybrid models (CNN + ViT) consistently outperformed their standalone CNN counterparts by leveraging both spatial and semantic features. The added attention mechanism from ViT modules enabled improved context understanding [6].
- Among the hybrid models, Inception + ViT achieved the highest test accuracy of 99%, demonstrating excellent generalization [6].
- DenseNet201 + ViT and EfficientNetV2-S + ViT achieved accuracies of 98.7% and 96.5%, respectively. The latter is particularly suited for real-time or resource-constrained environments [5].

Overall, this study confirms that hybrid architectures combining CNNs and ViTs significantly enhance classification performance for jute pest detection. The results underline the importance of advanced data augmentation techniques and model fusion in achieving both high accuracy and generalization across imbalanced multi-class datasets [6], [19].

From the comparative results, it is evident that hybrid models outperformed standalone CNNs in both overall accuracy and class-wise F1-score. The integration of ViTs helped the models better capture global image context, leading to

improved classification robustness—especially for visually similar pest categories. Inception+ViT showed the most stable and highest performance, establishing it as the best model in this study [6].

A. FUTURE WORK

While the proposed hybrid CNN + ViT models have demonstrated strong performance, several directions remain for further exploration and practical deployment:

- **Real-time Deployment:** Optimize hybrid models—especially EfficientNetV2-S + ViT—for edge and mobile applications in real-time pest detection [21].
- **Explainable AI:** Integrate interpretability tools like Grad-CAM or SHAP to visualize important regions influencing model predictions and improve user trust [1].
- **Multi-Modal Learning:** Enhance performance by incorporating additional inputs such as GPS location, weather conditions, or plant health metrics [21].
- **Active and Continual Learning:** Implement learning systems that adapt to new data or pest types over time, minimizing manual labeling efforts.
- **Synthetic Data Generation:** Use generative models like GANs to augment underrepresented pest classes and mitigate class imbalance [15].
- **Cross-Domain Generalization:** Extend the framework to other crop types and regions to assess its scalability across broader agricultural contexts [15].

These future directions can help transition the system from a research prototype into a practical, field-deployable tool for precision agriculture and pest management [21].

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CONTRIBUTIONS

This study offers the following significant contributions:

- **Development of a Hybrid CNN + ViT Model:** We present an advanced deep learning framework that combines Convolutional Neural Networks (CNNs) with Vision Transformers (ViTs) to improve the accuracy of jute pest classification by leveraging both detailed spatial and contextual global features.
- **We employed an existing dataset—the Jute Pest Dataset—and expanded its scope by adding a new pest class (Grasshopper) from a publicly accessible agricultural pest dataset.** This enhancement increased the dataset's diversity to 18 categories and addressed limitations associated with earlier datasets.
- **Comprehensive preprocessing was applied to the dataset to maintain balanced class representation and consistent**

data quality. The improved dataset structure supports dependable model training and evaluation for jute pest classification tasks.

These contributions provide a robust framework for designing effective and scalable solutions in intelligent agricultural pest detection.

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